

Chapter 2

Factor Models and Dimensionality Reduction

Appendix 1 - Why the Optimal Decoder Satisfies $W^{(1)} = \hat{P}A$

Giovanni Della Lunga
giovanni.dellalunga@unibo.it

Advanced Topics in Artificial Intelligence

Bologna - March, 2026

Roadmap

The Linear Autoencoder: Setup

We train a one-hidden-layer autoencoder with **linear** activation on the $N \times T$ matrix of demeaned returns $\bar{R}$.

After two preliminary steps (shown in the full proof):

- 1 The **bias terms** absorb the mean return $\bar{r}$ and drop out
- 2 The **optimal encoder** $W^{(0)}$ equals the pseudoinverse of the decoder $W^{(1)}$:

$$W^{(0)} = (W^{(1)'} W^{(1)})^{-1} W^{(1)'}$$

Substituting $W^{(0)}$ back, the optimization reduces to:

The Core Problem

$$\min_{W^{(1)}} \|\bar{R} - P_{W^{(1)}} \bar{R}\|_F^2$$

where $P_{W^{(1)}} = W^{(1)} (W^{(1)'} W^{(1)})^{-1} W^{(1)'}$ is the **orthogonal projector** onto the column space of $W^{(1)}$.

Goal of this presentation: show that the solution to this problem is

$$W^{(1)} = \hat{P}A$$

Tool 1 — The SVD of $\bar{R}$

The **Singular Value Decomposition** of $\bar{R} \in \mathbb{R}^{N \times T}$ (truncated to rank K) is:

$$\bar{R} = \hat{P} \hat{\Lambda} \hat{Q} + U$$

Object	Dimension	Property
$\hat{P}$	$N \times K$	orthonormal columns: $\hat{P}' \hat{P} = I_K$
$\hat{\Lambda}$	$K \times K$	diagonal, $\sigma_1 \geq \dots \geq \sigma_K > 0$
$\hat{Q}$	$K \times T$	orthonormal rows: $\hat{Q} \hat{Q}' = I_K$
U	$N \times T$	residual

Crucial orthogonality property

The residual U is **orthogonal to the column space of $\hat{P}$** :

$$\hat{P}' U = 0$$

This follows directly from the construction of the SVD.

Tool 2 — The Eckart–Young–Mirsky Theorem

Theorem (Eckart–Young–Mirsky, 1936)

Among all $N \times T$ matrices of rank at most K , the one closest to $\bar{R}$ in Frobenius norm is its truncated SVD:

$$\bar{R}^{(K)} = \hat{P}\hat{\Lambda}\hat{Q}$$

That is, for any matrix M with $\text{rank}(M) \leq K$:

$$\|\bar{R} - \hat{P}\hat{\Lambda}\hat{Q}\|_F \leq \|\bar{R} - M\|_F$$

Why does this apply here?

The reconstruction $P_{W^{(1)}}\bar{R}$ has rank at most K (the projector $P_{W^{(1)}}$ has rank K). So our minimization problem is exactly a *best rank- K approximation* problem:

$$\min_{W^{(1)}} \|\bar{R} - P_{W^{(1)}}\bar{R}\|_F^2 \iff \min_{M: \text{rank}(M) \leq K} \|\bar{R} - M\|_F^2$$

and Eckart–Young tells us the solution is $M^* = \hat{P}\hat{\Lambda}\hat{Q}$.

Tool 3 — Projectors and Subspaces

Key fact

An orthogonal projector P onto a subspace $\mathcal{S}$ is **uniquely determined by $\mathcal{S}$** , not by the particular basis chosen for $\mathcal{S}$.

Consequence: two matrices W and $\tilde{W}$ generate the **same projector** if and only if they have the **same column space**:

$$P_W = P_{\tilde{W}} \iff \text{col}(W) = \text{col}(\tilde{W})$$

Example

Let $\hat{P}$ have orthonormal columns and A be any $K \times K$ invertible matrix. Then $\text{col}(\hat{P}A) = \text{col}(\hat{P})$, because multiplying on the right by an invertible matrix is just a change of basis within the same subspace.

Therefore: $P_{\hat{P}A} = P_{\hat{P}} = \hat{P}\hat{P}'$.

Step 1 — Apply Eckart–Young

From the theorem, the **optimal reconstruction** is:

$$P_{W^{(1)}}^* \bar{R} = \hat{P} \hat{\Lambda} \hat{Q}$$

So we need to find $W^{(1)}$ such that:

$$P_{W^{(1)}} \bar{R} = \hat{P} \hat{\Lambda} \hat{Q}$$

The key rewriting: we claim that

Claim

$$\hat{P} \hat{\Lambda} \hat{Q} = \hat{P} \hat{P}' \bar{R}$$

If this claim is true, then the condition becomes:

$$P_{W^{(1)}} \bar{R} = \hat{P} \hat{P}' \bar{R}$$

which suggests that $P_{W^{(1)}} = \hat{P} \hat{P}'$, i.e. the optimal projector is the projection onto the column space of $\hat{P}$.

Let us prove the claim

Step 2 — Why $\hat{P}\hat{\Lambda}\hat{Q} = \hat{P}\hat{P}'\bar{R}$ (The Key Calculation)

Start from the SVD decomposition of $\bar{R}$:

$$\bar{R} = \underbrace{\hat{P}\hat{\Lambda}\hat{Q}}_{\text{rank-}K \text{ part}} + \underbrace{U}_{\text{residual}}$$

Apply $\hat{P}\hat{P}'$ to both sides:

$$\hat{P}\hat{P}'\bar{R} = \hat{P}\hat{P}'(\hat{P}\hat{\Lambda}\hat{Q}) + \hat{P}\hat{P}'U$$

First term:

$$\hat{P}\hat{P}'(\hat{P}\hat{\Lambda}\hat{Q}) = \hat{P}(\underbrace{\hat{P}'\hat{P}}_{=I_K})\hat{\Lambda}\hat{Q} = \hat{P}\hat{\Lambda}\hat{Q}$$

Second term:

$$\hat{P}\hat{P}'U = \hat{P}(\underbrace{\hat{P}'U}_{=0}) = 0$$

because $\hat{P}'U = 0$ (orthogonality property of the SVD).

Result

Step 3 — Identify the Optimal Projector

We have shown that the optimal reconstruction satisfies:

$$P_{W^{(1)}}^* \bar{R} = \hat{P} \hat{P}' \bar{R}$$

Does this mean $P_{W^{(1)}}^* = \hat{P} \hat{P}'$?

Yes — but we need to be careful. The equation $P_{W^{(1)}}^* \bar{R} = \hat{P} \hat{P}' \bar{R}$ must hold for *all* configurations of returns, not just for the specific $\bar{R}$ in our sample. Since the projector is a property of the model (not of the data), we can identify:

Optimal Projector

$$P_{W^{(1)}}^* = \hat{P} \hat{P}'$$

This is the orthogonal projector onto the **K-dimensional subspace spanned by the columns of $\hat{P}$** (the top- K left singular vectors of $\bar{R}$).

Step 4 — From the Projector to $W^{(1)}$

We need: which matrices $W^{(1)}$ generate the projector $\hat{P}\hat{P}'$?

Recall that $P_{W^{(1)}} = W^{(1)}(W^{(1)'}W^{(1)})^{-1}W^{(1)'}$ is the projector onto $\text{col}(W^{(1)})$. Since a projector uniquely determines its subspace:

$$P_{W^{(1)}} = \hat{P}\hat{P}' \iff \text{col}(W^{(1)}) = \text{col}(\hat{P})$$

Which $N \times K$ matrices have the same column space as $\hat{P}$?

Exactly those of the form $\hat{P}A$ for some **invertible** $K \times K$ matrix A :

- $\text{col}(\hat{P}A) = \text{col}(\hat{P})$ because A invertible $\Rightarrow A$ is a change of basis within $\mathbb{R}^K$
- Conversely, any matrix with $\text{col}(W^{(1)}) = \text{col}(\hat{P})$ can be written as $\hat{P}A$ for some invertible A

Conclusion

$$W^{(1)} = \hat{P}A \quad \text{for any invertible } K \times K \text{ matrix } A$$

The Rotation A Cancels Out

One might worry: if there are infinitely many solutions $W^{(1)} = \hat{P}A$, is the model ill-defined?

No — because A cancels in every economically relevant quantity.

Reconstruction:

$$P_{W^{(1)}} \bar{R} = \hat{P} \hat{P}' \bar{R} = \hat{P} \hat{\Lambda} \hat{Q} \quad (\text{independent of } A)$$

Optimal encoder (from Step 2 of the full proof):

$$W^{(0)} = (W^{(1)'} W^{(1)})^{-1} W^{(1)'} = A^{-1} \hat{P}'$$

Estimated factors:

$$\hat{F} = W^{(0)} \bar{R} = A^{-1} \hat{P}' \bar{R} = A^{-1} \hat{\Lambda} \hat{Q}$$

Fitted returns (A cancels explicitly):

$$W^{(1)} \hat{F} = (\hat{P}A)(A^{-1} \hat{\Lambda} \hat{Q}) = \hat{P} \hat{\Lambda} \hat{Q} \quad \checkmark$$

Analogy

Choosing A is like choosing Fahrenheit vs. Celsius: the temperature (the factor space) is the same; only the coordinate system changes.

Proof at a Glance

$$\text{Core problem: } \min_{W^{(1)}} \|\bar{R} - P_{W^{(1)}} \bar{R}\|_F^2$$

Theorem

$$\text{Eckart-Young: optimal reconstruction} = \hat{P} \hat{\Lambda} \hat{Q}$$

Step 2

$$\hat{P} \hat{\Lambda} \hat{Q} = \hat{P} \hat{P}' \bar{R} \quad (\text{since } \hat{P}' \hat{P} = I_K \text{ and } \hat{P}' U = 0)$$

Step 3

$$\text{Optimal projector: } P_{W^{(1)}}^* = \hat{P} \hat{P}'$$

Step 4

$$\text{col}(W^{(1)}) = \text{col}(\hat{P}) \implies W^{(1)} = \hat{P} A, \quad A \text{ invertible}$$

What is Uniquely Identified

Even though A is not uniquely determined, the following **are** uniquely identified:

Object	Formula	Unique?
K -dimensional factor subspace	$\text{col}(\hat{P})$	Yes
Optimal projector	$\hat{P}\hat{P}'$	Yes
Fitted returns	$\hat{P}\hat{\Lambda}\hat{Q}$	Yes
Residuals $\hat{u}_t$	$r_t - \hat{r}_t$	Yes
R^2 , loss function value	$\ U\ _F^2$	Yes
Individual loadings $W^{(1)}$	$\hat{P}A$	No (rotation free)
Individual factors $\hat{F}$	$A^{-1}\hat{\Lambda}\hat{Q}$	No (rotation free)

Bottom line

The linear autoencoder and PCA produce **identical reconstructions** and span the **same factor space**. The rotation A is a representation artifact, not an economic degree of freedom.